Chemistry GCE O Level 2022 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0515 Chemistry 1, June 2022, Ordinary Level). Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D. Non-programmable calculators are allowed. Relative atomic masses: Copper (Cu) = 64, Carbon (C) = 12.0, Hydrogen (H) = 1.0, Nitrogen (N) = 14.0 (the value for Oxygen (O) is not printed on the paper).

Multiple choice questions - choose the one correct answer, A, B, C or D (Questions 20-22 and 29-31 use special answer formats explained in the questions).

1. Which of the following is NOT a sub-atomic particle?

a) Proton
b) Neutron
c) Nucleon
d) Electron

2. Which of the following relights a glowing splint?

a) CO_2
b) NO_2
c) O_2
d) SO_2

3. Why is ammonia described as an alkaline gas?

a) It turns damped red Litmus paper blue
b) It forms white fumes with conc. $HCl_{(aq)}$
c) It has no effect on litmus paper
d) It contains a hydroxide ion

4. Identify a fertilizer made from Sulphuric acid

a) Potassium Chloride
b) Ammonium sulphate
c) Ammonium Nitrate
d) Ammonium Sulphite.

5. Which of the following substances causes hardness in water?

a) $NaHCO_3$
b) Na_2CO_3

c) $MgSO_4$
d) K_2SO_4

6. Determine the number of neutrons in the atom with a mass number of 35 and an atomic number of 17.

a) 37
b) 17
c) 18
d) 52

7. What is the function of Chlorine in the treatment of water?

a) To remove odour
b) To remove taste
c) To remove solid particles
d) To kill germs

8. Questions 8-9

A current of 1.5 A is passed for 1hour 30minutes during the purification of copper by electrolysis.

Which of the following reactions takes place at the anode?

a) $Cu_{(s)} \rightarrow Cu^{2+} + 2e^-$
b) $Cu^{2+} + 2e^- \rightarrow Cu_{(s)}$
c) $Cu^{2+} - 2e^- \rightarrow Cu_{(s)}$
d) $Cu_{(s)} + 2e^- \rightarrow Cu^{2-}$

9. Questions 8-9

A current of 1.5 A is passed for 1hour 30minutes during the purification of copper by electrolysis.

Calculate the mass of pure copper deposited at the Anode?

a) 0.045g
b) 2.70g
c) 0.00075g
d) 0.039g

10. Given the equation

$$2CH_3COOH + Na_2CO_3 \rightarrow 2X + CO_2 + H_2O$$

Identify the organic compound, X.

a) Sodium ethoxide
b) Sodium methanoate
c) Sodium hydroxide
d) Sodium ethanoate

11. Two elements M and P have atomic number 12 and 7 respectively. Give the formula of the compound formed between M and P

a) M_2P_3
b) MP_2
c) M_3P_2
d) M_5P_2

12. A soil sample has a pH of 4. The soil sample is

a) Basic
b) Acidic
c) Neutral
d) Alkaline

13. A hydrocarbon contains 85.7% C and 14.3% H. Determine the empirical formula of the hydrocarbon.

a) CH_2
b) CH
c) CH_3
d) C_2H_4

14. When a few drops of aqueous silver nitrate is added to a salt solution, a yellow precipitate is formed. Identify the anion in the salt.

a) Fluoride ion
b) Chloride ion
c) Bromide ion
d) Iodide ion

15. State the function of Super-heated water in the extraction of sulphur.

a) To separate the sulphur
b) To dissolve sulphur
c) To melt sulphur
d) To pull up the sulphur

16. Given the equation

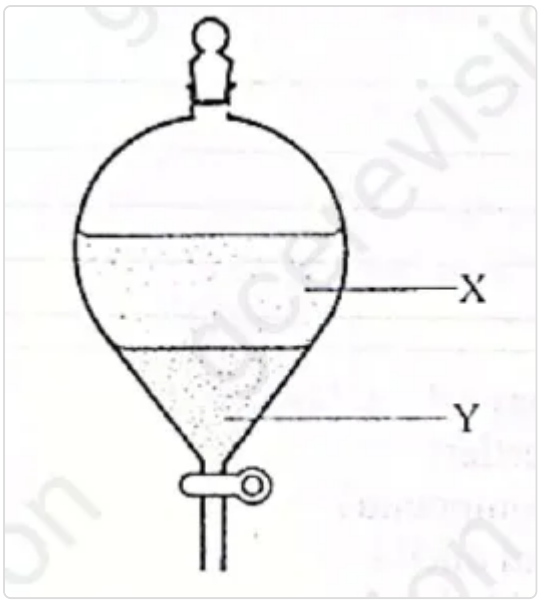
$$CuO_{(s)} + H_{2(g)} \rightarrow Cu_{(s)} + H_2O_{(g)}$$

Select from the equation, the oxidizing agent.

a) H_2
b) CuO
c) Cu
d) H_2O

17. Questions 17-18

The following diagram represents a separation technique used to separate a mixture of X and Y

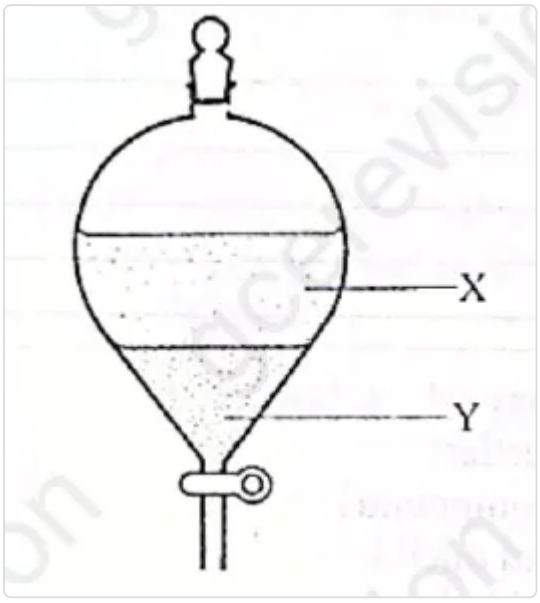


Identify a pair of substances that could be X and Y respectively.

a) ethanol and petrol
b) Petrol and water
c) Petrol and kerosene
d) Water and petrol

18. Questions 17-18

The following diagram represents a separation technique used to separate a mixture of X and Y



State the property of X and Y that makes it suitable for separation by this method.

a) Difference in volatility
b) Difference in melting points
c) Difference in boiling points
d) Difference in density

19. Arrange the elements; Calcium, Iron, Sodium and Zinc in order of increasing reactivity, starting from the least reactive.

a) Sodium, Zinc, Iron, Calcium
b) Zinc, Iron, Calcium, Sodium

c) Iron, Zinc, Sodium, Calcium
d) Iron, Zinc, Calcium, Sodium

20. Questions 20-22

Instructions (Questions 20-22): each of the following questions consists of a statement in the left-hand column followed by a second statement in the right-hand column. Decide whether each of the statements is TRUE or FALSE. Then choose A, B, C or D.

First statement	Second statement
Alkali metals are very good reducing agents	The alkali metals are easily oxidized

a) If both statements are true and the second statement is a correct explanation of the first statement
b) If both statements are true but the second statement is not a correct explanation of the first statement
c) If the first statement is true but the second statement is false
d) If the first statement is false but the second statement is true

21. Questions 20-22

Instructions (Questions 20-22): each of the following questions consists of a statement in the left-hand column followed by a second statement in the right-hand column. Decide whether each of the statements is TRUE or FALSE. Then choose A, B, C or D.

First statement	Second statement
C_2H_2 and C_6H_6 have the same chemical and physical properties	C_2H_2 and C_6H_6 have the same percentage by mass of hydrogen

a) If both statements are true and the second statement is a correct explanation of the first statement
b) If both statements are true but the second statement is not a correct explanation of the first statement
c) If the first statement is true but the second statement is false
d) If the first statement is false but the second statement is true

22. Questions 20-22

Instructions (Questions 20-22): each of the following questions consists of a statement in the left-hand column followed by a second statement in the right-hand column. Decide whether each of the statements is TRUE or FALSE. Then choose A, B, C or D.

First statement	Second statement
Acid/base indicators are dyes	Indicators develop colour only in the presence of a strong acid or strong base.

a) If both statements are true and the second statement is a correct explanation of the first statement
b) If both statements are true but the second statement is not a correct explanation of the first statement
c) If the first statement is true but the second statement is false
d) If the first statement is false but the second statement is true

23. Which of the following carbonates is stable on heating?

a) Lithium carbonate
b) Sodium carbonate

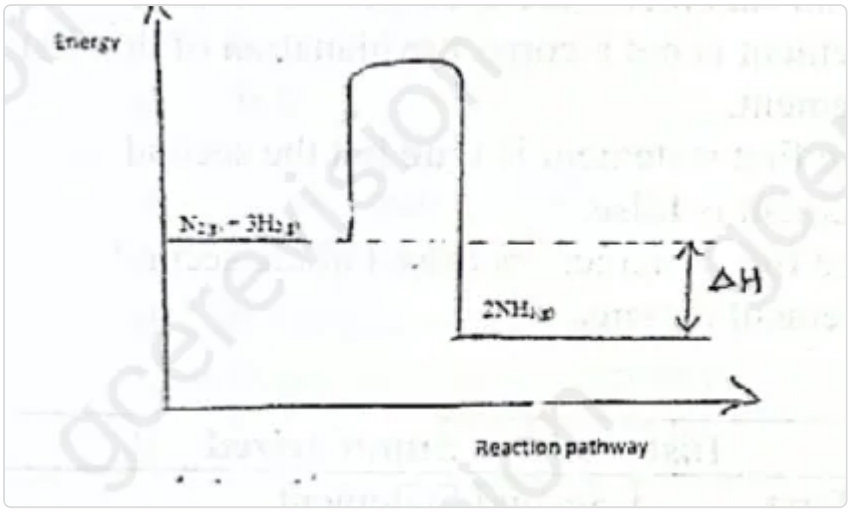
c) Ammonium carbonate
d) Calcium carbonate

24. Which of the following will decolourise bromine water?

a) Ethene
b) Ethane
c) Butane
d) Ethanol

25. Questions 25-26

Consider the energy level diagram below for a reversible reaction:

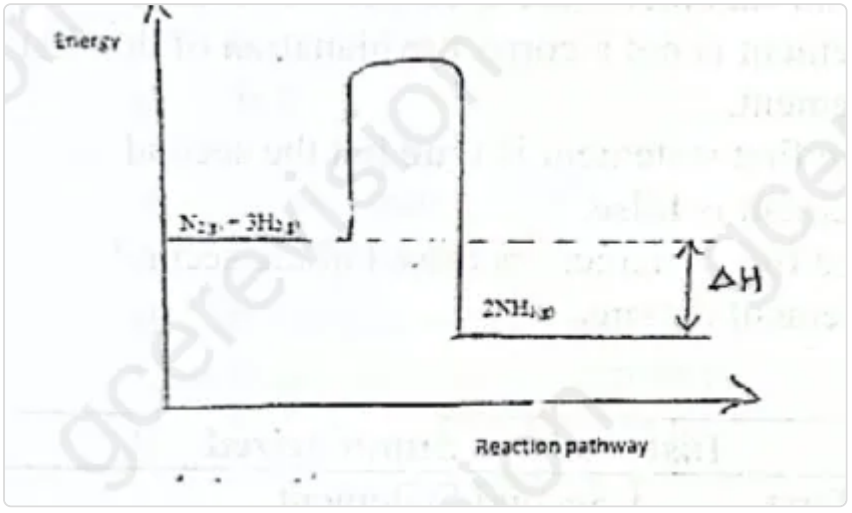


What heat change occurs in the reaction?

a) Heat is absorbed during the reaction
b) Endothermic reaction occurs
c) Heat is released
d) Activation energy increases

26. Questions 25-26

Consider the energy level diagram below for a reversible reaction:



What is the effect of an increase in pressure on the equilibrium position?

a) Equilibrium shifts to the right
b) Equilibrium shifts to the left
c) Equilibrium position is unaffected
d) Equilibrium position shifts both to the right and left

27. What is the percentage composition of oxygen in air

a) 78%
b) 21%
c) 0.03%
d) 1%

28. Calculate the percentage composition by mass of Nitrogen in Ammonium phosphate $(NH_4)_3PO_4$. (RMM = 149)

a) 28.2%
b) 9.4%
c) 16.1%
d) 20.8%

29. Questions 29-31

Instructions (Questions 29-31): for each of the questions, ONE or MORE of the response(s) given is (are) correct. Decide which of the response(s) is (are) correct. Then choose: A. If 1, 2 and 3 are correct; B. If 1 and 3 are correct; C. If 2 and 4 are correct; D. If 4 only is correct.

Which of the following substances is(are) allotropic?

1. Carbon
2. Sulphur
3. Phosphorus
4. Calcium Carbonate

a) 1, 2 and 3 only
b) 1 and 3 only
c) 2 and 4 only
d) 4 only

30. Questions 29-31

Instructions (Questions 29-31): for each of the questions, ONE or MORE of the response(s) given is (are) correct. Decide which of the response(s) is (are) correct. Then choose: A. If 1, 2 and 3 are correct; B. If 1 and 3 are correct; C. If 2 and 4 are correct; D. If 4 only is correct.

Transition metals have the following characteristic properties:

1. Form coloured compounds
2. Are malleable and ductile
3. Have variable oxidation states
4. Conduct electricity

Options as printed: A) 1, 2 and 3 only B) 1 and 3 only C) 2 and 4 only D) 4 only

31. Questions 29-31

Instructions (Questions 29-31): for each of the questions, ONE or MORE of the response(s) given is (are) correct. Decide which of the response(s) is (are) correct. Then choose: A. If 1, 2 and 3 are correct; B. If 1 and 3 are correct; C. If 2 and 4 are correct; D. If 4 only is correct.

Substance X reacts with dilute H_2SO_4 producing copper(II)sulphate salt. Identify X

1. Copper metal
2. Copper oxide
3. Copper hydroxide
4. Copper carbonate

Options as printed: A) 1, 2 and 3 only B) 1 and 3 only C) 2 and 4 only D) 4 only

32. Why does powdering marble chips increase the rate of the following reaction;

$$CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$$

a) The total surface area reduces
b) Collision frequency increases
c) The number of effective collision decreases
d) The rate increases with less contact of marble and acid

33. Which of the following is a reduction reaction?

a) $Cu_{(s)} \rightarrow Cu^{2+}_{(aq)} + 2e^-$
b) $4OH^-_{(aq)} \rightarrow 2H_2O_{(l)} + O_{2(g)} + 4e^-$
c) $2Cl^-_{(aq)} \rightarrow Cl_{2(g)} + 2e^-$
d) $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$

34. How is chlorine collected in the laboratory?

a) Downward delivery
b) Over water
c) Upward delivery
d) Downward displacement of air

35. $25\ cm^3$ of $0.1\ M\ NaOH$ neutralizes $20\ cm^3$ of H_2SO_4 solution. Calculate the concentration of H_2SO_4

a) 0.125M
b) 0.04M
c) 0.08M
d) 0.0625M

36. Identify the oxidizing agent from the equation;

$$2H_2S_{(g)} + SO_{2(g)} \rightarrow 2H_2O_{(l)} + 3S_{(s)}$$

a) H_2S
b) SO_2
c) H_2O
d) S

37. A mixture is:

a) A substance that contains two or more substances not chemically combined
b) A substance formed when two or more substances combine chemically through bond formation.
c) A substance that cannot be broken down into simpler units
d) A substance that has weight and occupies space.

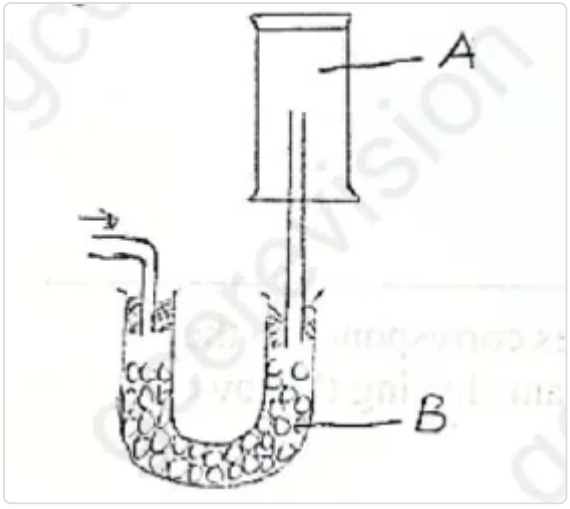
38. Elements A and B have atomic numbers 12 and 16 respectively. Identify the type of bonding formed between them

a) Covalent bonding
b) Metallic bonding
c) Covalent bonding
d) Ionic bonding

39. From the electronic configurations, which of the following belong to the same period?

a) 2,8,8,6 and 2,8,8
b) 2,8 and 2,8,3
c) 2,8,1 and 2,8,3
d) 2,8,1 and 2,8,8,1

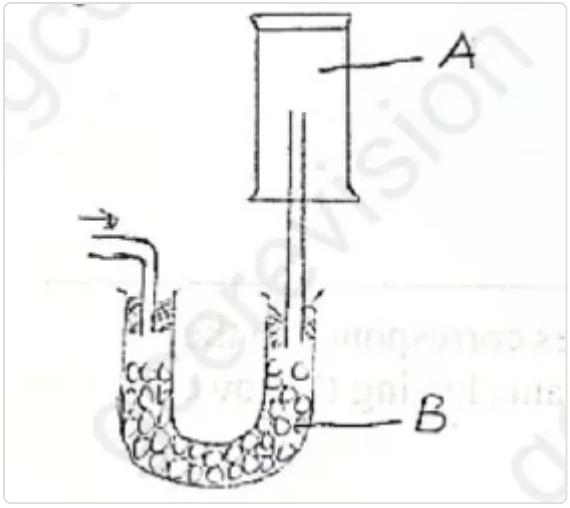
40. Questions 40-41 concern the following diagram:



Identify the gas, A, that can be dried and collected in this way.

a) CO_2
b) NH_3
c) Cl_2
d) SO_2

41. Questions 40-41 concern the following diagram:



Identify the drying agent 'B'

a) Quicklime
b) Slaked lime
c) Calcium chloride
d) Sulphuric acid

42. Which of the following chemical equations is balanced?

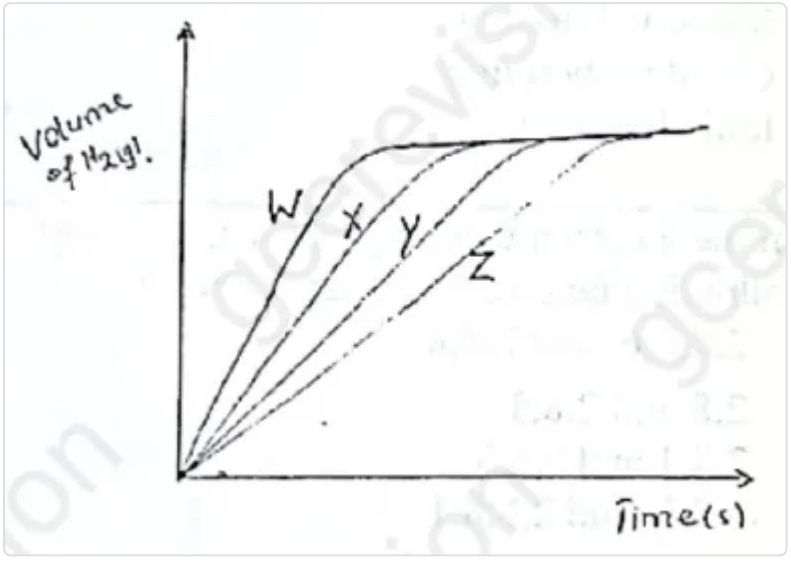
a) $\mathrm{Na_2CO_{3(aq)} + HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}}$
b) $\mathrm{3Fe_{(s)} + 4H_2O_{(l)} \rightarrow Fe_3O_{4(s)} + H_2O_{(l)}}$
c) $\mathrm{NO_{(g)} + O_{2(g)} \rightarrow NO_{2(g)}}$
d) $\mathrm{Mg_{(s)} + 2HNO_{3(aq)} \rightarrow Mg(NO_3)_{2(aq)} + H_{2(g)}}$

43. A salt which when exposed to air forms a solution is referred to as:

a) Deliquescent
b) Efflorescent
c) Hygroscopic
d) Hydrated

44. Questions 44-45

The diagram shows the rate of a reaction at different concentrations.

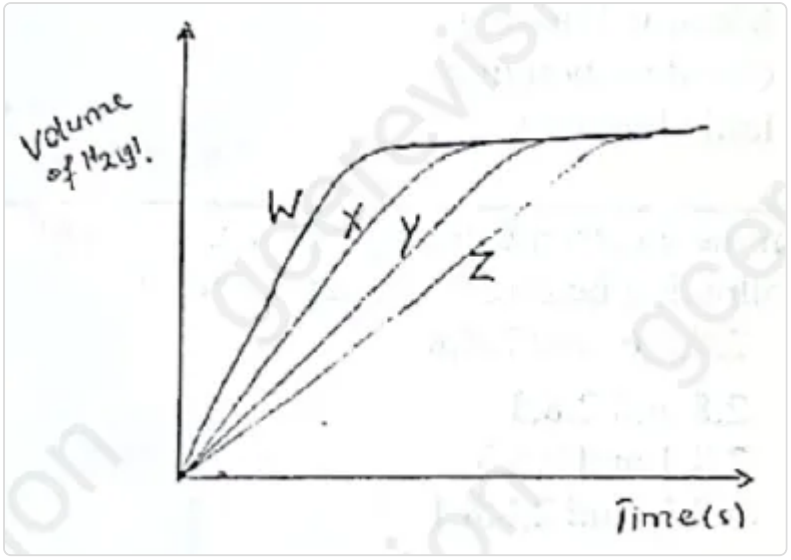


Which of the curves shows the fastest reaction rate?

a) Y
b) X
c) W
d) Z

45. Questions 44-45

The diagram shows the rate of a reaction at different concentrations.



Which of the curves corresponds to the reaction with reactants having the lowest concentration?

a) X
b) W
c) Z
d) Y

46. Identify the product formed when hexan-1,6-diamine reacts with hexanedioic acid

a) Styrene
b) Terylene
c) Perspex
d) Nylon6,6

47. Which gas is always given off when a metal nitrate is heated?

a) O_2
b) NO_2
c) NO
d) NH_3

48. A current of 2A was passed for 90 minutes through an electrolytic cell. Calculate the quantity of electricity in coulombs.

a) 180C
b) 7200C
c) 3600C
d) 10800C

49. What is the charge on a proton?

a) Negative
b) Neutral
c) Positive
d) No charge

50. The change of state from solid to liquid is known as

a) Condensation
b) Melting
c) Freezing
d) Sublimation

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

View the correction

Scan me

edukora.africa - Free-access paper, feel free to share. The correction is for subscribers only.